

8DM60: Multimodal AI in Biomedicine

Lecture 3: Guided self-study (BZ)

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Problem 1

scVI adapts a variational autoencoder architecture specifically to model the characteristics of single-cell RNA-seq data.

Explain how scVI's architecture and probabilistic generative formulation explicitly incorporate structural components to address each of the following five key challenges. For each, identify the relevant component of scVI and explain how it helps:

1. **Sparsity and excess zeros**
2. **Varying sequencing depth (library size)**
3. **Technical batch effects**
4. **Transcriptional / biological noise**
5. **Scalability to large datasets (millions of cells)**

For each challenge, identify the relevant component of scVI and explain how it helps.

Problem 2

A researcher converts each cell into a list of genes and feeds it into a standard Transformer exactly as if it were a sentence, using only gene identities and positional embeddings.

Why is this representation inappropriate for scRNA-seq data? Describe two changes needed to make the input representation biologically meaningful.

Problem 3

When single-cell count data are passed through scGPT, each gene i starts with an input embedding

$$h_i = \mathrm{emb}_g(g_i) + \mathrm{emb}_x(x_i),$$

combining gene identity and expression value. After several self-attention layers, the model produces a contextualized representation for each gene.

Consider TP53 in a resting T cell and in a stressed cancer cell. Assume TP53 has the same expression value in both cells.

Will the initial input embedding of TP53 be the same or different between the two cells? Will its post-Transformer representation be the same or different? Explain what biological capability self-attention provides.

Problem 4

A laboratory generates a new unlabeled scRNA-seq dataset across four batches and compares:

A: zero-shot embeddings from a frozen foundation model (e.g. scGPT)
B: a model fitted directly to the new dataset (e.g. scVI)

Based on the zero-shot benchmarks discussed in Kedzierska et al., how would you expect the two strategies to compare in terms of batch integration and biological conservation? Explain.

Now suppose batch is strongly confounded with biology (e.g. cancer and normal cells are measured in different batches). Could the zero-shot foundation-model representation now have an advantage over aggressive batch integration? Explain.

Problem 5

A research team uses CPA (Compositional Perturbation Autoencoder) to model cellular drug responses. They want to predict the transcriptomic response of cells exposed to a completely novel small molecule, Drug X, that was never included in the training dataset.

Can standard CPA generate a meaningful zero-shot prediction for Drug X? Explain why or why not based on CPA's perturbation representation.

If the loss weight of CPA's adversarial perturbation discriminator is set to zero during training, how would this affect the learned basal state (z_{basal}) and the model's ability to predict counterfactual drug combinations?

Problem 6

GEARS incorporates two distinct graph neural networks operating on two prior biological graphs: a gene co-expression graph G_{gene} and a Gene Ontology-derived perturbation graph G_{pert}.

A student suggests removing G_{pert} and instead using G_{gene} to perform message passing for perturbation embeddings. Why would this undermine GEARS' ability to predict the transcriptomic effect of knocking out a gene that was never experimentally perturbed during training?

Explain the conceptual difference between the biological relationships captured by G_{gene} and G_{pert}.

Problem 7

A deep neural network predicts how the expression of $G = 10{,}000$ genes across the transcriptome changes under different treatment conditions. On a held-out test set, it achieves a low global mean squared error, $\mathrm{MSE} = 0.05$. A simple No-Change baseline, which predicts the unperturbed control expression state for every treatment, achieves $\mathrm{MSE} = 0.06$.

For each treatment, only about 50 of the 10,000 genes actually change expression substantially.

1. Why can the No-Change baseline achieve such a low global error?
2. Why can global MSE give an overly optimistic impression of how well the neural network predicts treatment-specific effects?
3. How should the evaluation metric or feature selection be modified to properly measure true treatment effects?